COMSATS University Islamabad, Lahore Campus

Midterm Examination – Spring 2024

Course Title:	Web Technologies				Course Code:	CSC330	Credit Hours: 4(3,1)
Instructor/s:	M. Ali Hassan				Program Name:	BS	
Semester:		Batch:		Section:		Date:	24-04-2024
Time Allowed:	1.5 Hours				Maximum Marks:		50
Student's Name:					Reg. No.		

Question 1: Answer the following questions [5*3=15]

CLO-1; Bloom Taxonomy Level: Understanding

1. Explain the different parts of the CSS box model (content, padding, border, margin) and how they contribute to the overall dimensions and positioning of an element on the web page.
2. What is the Document Object Model (DOM) in web development? How does JavaScript interact with the DOM?
3. What are HTTP methods and their respective purposes in web development?
4. Explain the concept of responsive web design. Why is it important in today's web development?
5. What is the difference between require() and import() in JavaScript?

Question 2: Given the following HTML structure for an image gallery: [5]

CLO 2; Bloom Taxonomy Level: Applying

```
<div class="gallery">
  <img src="image1.jpg" alt="Image 1">
  <img src="image2.jpg" alt="Image 2">
  <img src="image3.jpg" alt="Image 3">
</div>
```

Write CSS to style the gallery with the following specifications:

- Display the images in a grid layout with 3 columns.
- Add a margin of 10px around each image

Question 3: Given the following HTML structure for a navigation menu: [5]

CLO 2; Bloom Taxonomy Level: Applying

```
<nav class="main-menu">
  <ul>
    <li><a href="#">Home</a></li>
    <li><a href="#">About</a></li>
    <li><a href="#">Services</a></li>
```

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```
    <li><a href="#">Contact</a></li>
  </ul>
</nav>
```

Write CSS to style the navigation menu with the following specifications:

- Display the menu items horizontally.
- Add a margin of 15px around each menu item.
- Set a background color of #333333 to the menu items
- Change the text color of the menu items to #ffffff.
- Apply a hover effect that changes the background color to #555555.

Question 4: You are given an HTML <select> element with options representing colors. Write JavaScript code to change the background color of the web page to the selected color when an option is chosen. [5]

CLO-2; Bloom Taxonomy Level: Applying

```
<select id="colorSelect" onchange="changeBackgroundColor()">
  <option value="red">Red</option>
  <option value="blue">Blue</option>
  <option value="green">Green</option>
  <option value="yellow">Yellow</option>
</select>
```

Question 5: Given the following HTML structure for a book library: [20]

CLO-2; Bloom Taxonomy Level: Applying

```
<div id="book-library">
  <input type="text" id="title" placeholder="Title">
  <input type="text" id="author" placeholder="Author">
  <button id="add-book">Add Book</button>
  <ul id="books">
    <!-- Books will be added here -->
  </ul>
</div>
```

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Write jQuery code to implement CRUD operations for books:

- Create: Add a new book to the library when the "Add Book" button is clicked.
- Read: Display all books from an array when the page loads.
- Update: Allow updating a book's details by clicking on it.
- Delete: Remove a book when its delete button is clicked.